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Dynamic Pseudo-Temporal Prompt Generation for Accurate Few-Shot Remote Sensing Change Detection

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***Abstract*—Deep learning-based remote sensing change detection (RSCD) methods have achieved remarkable progress by exploiting deep neural networks to model temporal semantic discrepancies from sufficient labeled bi-temporal image pairs. However, these methods suffer from significant performance degradation in few-shot scenarios, since scarce labeled data severely weakens the model's capacity for temporal difference modeling and degrades model generalization. To mitigate this issue, this study proposes a novel dynamic pseudo-temporal prompt generation-enhanced framework for few-shot RSCD. Specifically, we design a dynamic pseudo-temporal prompt generation module that adaptively produces sample-specific transitional prompts according to the change-aware differences between input bi-temporal images, thereby avoiding explicit pixel-level intermediate image generation. Subsequently, the original bi-temporal images together with the generated transitional prompt are fed into a pre-trained video encoder, which enables the model to simulate implicit three-phase temporal evolution and capture comprehensive temporal semantic variations throughout the whole change process. Furthermore, a multi-branch temporal decoder is introduced to decode complementary temporal differences from the constructed pseudo-temporal representation. To effectively optimize the complementary predictions produced by the multi-branch temporal decoder, we formulate a mixed pseudo-temporal transitional loss. This loss jointly supervises the main endpoint change prediction, the progressive intermediate prediction, and the adjacent transition prediction, enabling the network to learn more reliable temporal change representations under few-shot supervision. Extensive experiments on public RSCD benchmarks demonstrate that the proposed method consistently outperforms recent state-of-the-art approaches across diverse few-shot settings.**

***Index Terms*—Remote sensing change detection, few-shot learning, pseudo-temporal transitional prompt, mixed pseudo-temporal transitional loss.**

I. Introduction

REMOTE sensing change detection (RSCD) aims to identify land-cover or land-use changes by comparing images acquired over the same geographical area at different times. It is a fundamental task in remote sensing image analysis and has been widely applied to urban expansion monitoring [1], disaster damage assessment [2], agricultural land-use change tracking [3], and environmental monitoring. With the increasing availability of high-resolution remote sensing imagery, accurate RSCD is becoming increasingly important for understanding dynamic surface evolution in complex real-world scenarios.

In recent years, deep learning-based RSCD methods have achieved remarkable progress by learning discriminative change representations from labeled bi-temporal image pairs. Representative methods, including CNN-based Siamese networks [4]–[9], attention-based models [10], Transformer architectures [11]–[16], and visual foundation model-based approaches [17]–[19], have been widely explored to enhance bi-temporal feature interaction and semantic change representation. Although these methods achieve promising performance under fully supervised settings, they usually rely on abundant dense annotations. When only a few annotated samples are available, their performance often degrades significantly due to overfitting, insufficient temporal difference modeling, and poor generalization to unseen change patterns [20], [21].

To alleviate the dependence on dense annotations, existing few-shot or limited-data RSCD methods mainly enhance feature representation through few-shot learning strategies [22], modality-specific few-shot modeling [23], cross-domain or cross-scene information [24], or data-driven change simulation [25], [26]. In addition, some methods enrich supervision signals through data augmentation [27], generated samples [25], [26], pseudo-label learning [28], [29], semi-supervised learning [20], [21], or self-supervised objectives [30]. These strategies can partially reduce annotation requirements, but their effectiveness is often constrained by external knowledge reliability, domain gaps, generated sample quality, or pseudo-label accuracy. More importantly, most existing methods still focus on enhancing feature representation or expanding supervision signals, while the intrinsic pseudo-temporal transition information implied by each bi-temporal sample remains insufficiently explored.

In this paper, we argue that bi-temporal images not only reflect the endpoint discrepancy between pre-change and post-change observations, but also implicitly contain a latent transition process. However, most existing RSCD methods directly model endpoint differences, which compresses complex temporal evolution into a single-step comparison and becomes unreliable under few-shot supervision. Although recent video-inspired methods introduce learnable perception frames to enhance temporal modeling, such static frames are weakly correlated with sample-specific change content [31]. Meanwhile, explicitly generating pixel-level pseudo-temporal images may introduce unreliable visual details when annotations are scarce.

Based on these observations, building upon our preliminary

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conference work [32], we further propose a dynamic pseudo-temporal prompt generation-enhanced framework for few-shot RSCD. In this work, sample-specific pseudo-temporal transitional prompts are adaptively generated to implicitly represent the latent transition cues between the pre-change and post-change observations, and the preliminary framework is substantially extended from three aspects. First, the original single-decoder prediction framework is extended to a Multi-Branch Temporal Decoding Module (MTDM), which models endpoint changes, progressive cumulative changes, and adjacent transition changes from three complementary temporal perspectives. Second, based on the original temporal consistency supervision, we further develop a mixed pseudo-temporal transitional loss, which jointly incorporates main change supervision, progressive pseudo-label supervision, and temporal consistency supervision to collaboratively optimize the multi-branch predictions, thereby improving temporal representation learning and model generalization under limited annotations. Finally, more comprehensive experiments are conducted on multiple public RSCD benchmark datasets under various limited-data settings to validate the effectiveness and robustness of the proposed method.

The main contributions of this paper are summarized as follows:

- We propose a dynamic pseudo-temporal prompt generation framework for few-shot RSCD, which adaptively produces sample-specific transitional prompts to mine intrinsic transition cues while avoiding explicit pixel-level intermediate image generation.
- We develop a multi-branch temporal decoding module to exploit pseudo-temporal change cues from complementary perspectives, including endpoint changes, progressive cumulative changes, and adjacent transition changes.
- We formulate a mixed pseudo-temporal transitional loss by combining main supervision, progressive pseudo-label supervision, and temporal consistency supervision to improve the reliability of temporal change representation learning.

Extensive experiments on public RSCD benchmarks demonstrate that the proposed method consistently outperforms recent state-of-the-art approaches across diverse few-shot settings.

II. Related Work

In this section, we first review mainstream remote sensing change detection methods, and then discuss few-shot change detection methods closely related to our work.

A. Mainstream Remote Sensing Change Detection Methods

Early remote sensing change detection methods mainly relied on image differencing, spectral-index analysis, handcrafted descriptors, and thresholding strategies [33], [34]. With the development of deep learning, change detection has gradually shifted toward an end-to-end feature learning paradigm. CNN-based methods usually model bi-temporal relationships through early fusion or Siamese feature extraction. For example, FC-EF concatenates bi-temporal images at the image level, while FC-Siam-conc and FC-Siam-diff employ shared-weight Siamese encoders and fuse bi-temporal features by concatenation or differencing [4]. Following this paradigm, SNUNet-CD improves multi-scale feature fusion through dense skip connections [5], while TCDNet introduces a trilateral branch design to enhance the discrimination between changed and unchanged regions [35]. FarmCD further incorporates agricultural information into a gated network to enhance farmland change detection [36]. Other methods further improve change detection from the perspectives of boundary refinement, context aggregation, and bi-temporal feature interaction [6]–[9]. However, CNN-based methods mainly rely on local convolutional operations, which limits their ability to model long-range contextual dependencies and global change patterns.

To enhance global relationship modeling, attention mechanisms and Transformers have been widely introduced into remote sensing change detection. STANet strengthens bi-temporal feature interaction through a spatial attention mechanism [10], while BIT converts bi-temporal features into semantic tokens and leverages Transformers to model global contextual relationships [15]. Subsequently, SwinSUNet and ChangeFormer employ hierarchical Transformer architectures to extract multi-scale change representations, achieving promising performance under fully supervised settings [11], [12]. Recently, visual foundation models represented by the Segment Anything Model have also been introduced into change detection to exploit general visual priors and segmentation capabilities from large-scale pretrained models [17]–[19], [37]. Beyond change detection, related studies have investigated optimization-guided and physics-guided representation learning in remote sensing [38], [39]. Although these methods enhance bi-temporal feature interaction and semantic representation, most of them still focus on the final discrepancy between two temporal observations, compressing temporal evolution into a single-step change representation.

To alleviate this limitation, video-based change detection methods attempt to introduce temporal structures between bi-temporal images. Recently, Change3D regards pre-change and post-change images as an extremely short video and inserts learnable perception frames between them, enabling the video encoder to perform spatio-temporal feature interaction [31]. However, these perception frames are essentially static learnable representations and do not explicitly depend on the change content of each input sample, making it difficult to adaptively characterize the latent transition process across different scenes.

B. Few-Shot Remote Sensing Change Detection Methods

Existing few-shot change detection studies mainly focus on leveraging external knowledge or enriching supervision signals. One group of methods improves limited-data learning through prior knowledge, task adaptation, transfer learning, or cross-domain generalization. Some methods introduce few-shot learning into bi-temporal change detection and incorporate uncertainty estimation to improve prediction reliability in low-shot settings [22]. Transfer learning and cross-domain

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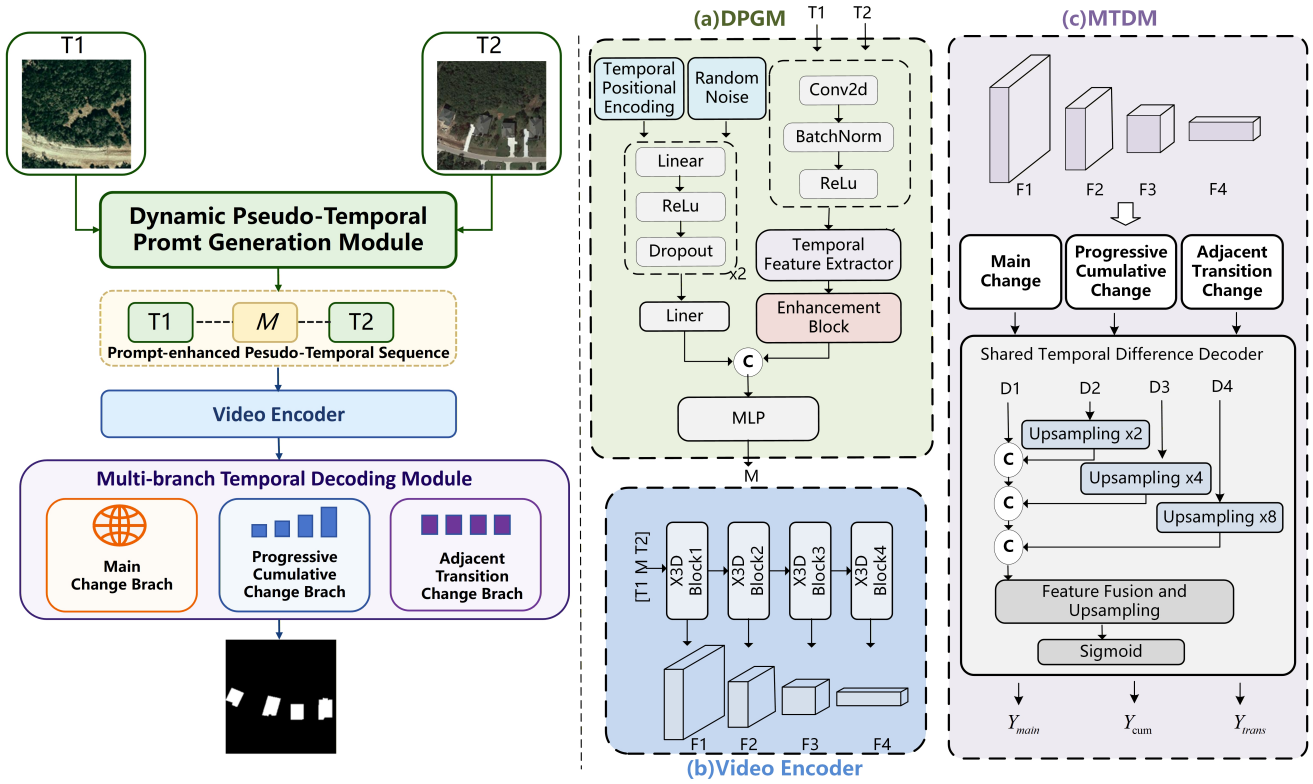

Fig. 1. Overall architecture of the proposed framework. The model consists of three main components: (a) a Dynamic Pseudo-Temporal Prompt Generation Module (DPGM), which generates a sample-specific pseudo-temporal transitional prompt according to the bi-temporal change content; (b) a pre-trained video encoder, which extracts multi-scale spatio-temporal features from the constructed pseudo-temporal representation; and (c) a Multi-Branch Temporal Decoding Module (MTDM), which models changed regions through the main change branch, progressive cumulative change branch, and adjacent transition change branch.

methods leverage source-domain data, pretrained models, or cross-scene knowledge to enhance adaptation when only a few labeled target samples are available [25]–[27]. For specific modalities such as hyperspectral images, 3D convolution and cross-scene collaborative learning have also been employed to model spectral-spatial structures under limited-sample conditions [23], [24]. Although these methods alleviate annotation scarcity, their performance often depends on source-target domain correlation, pretrained representation adaptability, or modality-specific structural priors.

Another group of methods mitigates insufficient supervision by expanding available training signals. Data augmentation and generative methods construct additional change samples through image fusion, change-region synthesis, or generative models. Semi-supervised methods exploit unlabeled bi-temporal images through pseudo-labels or consistency constraints [28], [29], while self-supervised methods learn transferable representations from unlabeled remote sensing images via pretext tasks or contrastive learning [30]. These methods increase supervision sources through additional samples, unlabeled data, or auxiliary objectives, but their effectiveness can be affected by generated sample quality, pseudo-label reliability, and pretraining-task suitability.

Although existing few-shot change detection methods alleviate annotation scarcity from different perspectives, they usually treat bi-temporal images as two static observations for direct comparison, leaving the latent temporal relationship between them insufficiently explored. In fact, bi-temporal images not only reflect the final difference between pre-change and post-change states, but also imply a latent evolution process from the initial state to the final state. Therefore, how to exploit intrinsic temporal cues from bi-temporal samples themselves under limited supervision remains an important problem for few-shot remote sensing change detection.

III. Methodology

A. Overall Framework

This paper proposes a dynamic pseudo-temporal prompt generation-enhanced framework for few-shot remote sensing change detection. Unlike traditional methods that directly compare bi-temporal images, the proposed method reformulates bi-temporal change detection as a short-term temporal modeling problem. As shown in Fig. 1, a sample-specific pseudo-temporal transitional prompt is first adaptively generated by the Dynamic Pseudo-Temporal Prompt Generation Module (DPGM) according to the input bi-temporal images. The generated prompt implicitly represents latent transition cues between the pre-change and post-change observations, thereby avoiding the difficulty of generating explicit pixel-level intermediate images. Subsequently, the original bi-temporal images and the generated transitional prompt are fed into a pre-trained video encoder to extract multi-scale spatio-temporal

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features. Finally, the Multi-Branch Temporal Decoding Module (MTDM) is introduced to produce change maps from three complementary branches, including the main change branch, the progressive cumulative change branch, and the adjacent transition change branch.

B. Dynamic Pseudo-Temporal Prompt Generation Module

To overcome the limited sample adaptability of static perception frames, this paper designs a Dynamic Pseudo-Temporal Prompt Generation Module (DPGM). Given the pre-change image T_1 and the post-change image T_2, DPGM adaptively generates a sample-specific pseudo-temporal transitional prompt according to the feature differences between the two temporal images. The generated prompt serves as an implicit pseudo-temporal representation for change perception, helping the video encoder exploit latent transition cues between the two observations while avoiding explicit pixel-level intermediate image generation.

Specifically, a shared encoder is first utilized to extract the basic features of the two temporal images:

$$H_1 = \phi(T_1), \quad H_2 = \phi(T_2), \tag{1}$$

where $\phi(\cdot)$ represents the shared spatial encoder composed of Conv2d, BN, and ReLU. H_1 and H_2 denote the extracted features of the pre-change and post-change images, respectively. To describe the differences between the two temporal phases, a temporal feature extraction module is designed to construct two types of difference features:

$$D_{\text{abs}} = |H_2 - H_1|, \quad D_{\text{dir}} = \text{ReLU}(H_2 - H_1). \tag{2}$$

Here, D_{abs} is used to characterize the magnitude of the change, while D_{dir} is used to preserve the directional change from T_1 to T_2. Subsequently, the two types of difference features are aggregated, and a global change description is obtained through a difference enhancement module:

$$F_d = \gamma\left(\frac{\text{GAP}(D_{\text{abs}}) + \text{GAP}(D_{\text{dir}})}{2}\right), \tag{3}$$

where $\text{GAP}(\cdot)$ denotes global average pooling and $\gamma(\cdot)$ represents the difference enhancement module. In this paper, $\gamma(\cdot)$ consists of two consecutive MLP–GroupNorm–ReLU–Dropout modules. Each MLP block contains fully connected layers for feature transformation, followed by GroupNorm, ReLU activation, and dropout regularization. The number of groups in GroupNorm is set to 8, and the dropout rate is set to 0.1. This module is designed to enhance change difference representations and improve feature robustness under limited annotation conditions.

To generate a dynamic pseudo-temporal prompt representation, this paper introduces a learnable noise base $N_{\text{base}} \in \mathbb{R}^{1\times 32}$ and maps it into a temporal noise feature:

$$N_{\text{temp}} = \varphi(N_{\text{base}}), \tag{4}$$

where $\varphi(\cdot)$ represents the temporal noise mapping network composed of Linear, ReLU, and Dropout modules. Finally, the change difference feature and the temporal noise feature are fused, and a sample-specific pseudo-temporal transitional prompt is generated through an MLP decoder:

$$M = \text{MLP}(\text{Concat}(N_{\text{temp}}, F_d)). \tag{5}$$

Here, M denotes the generated pseudo-temporal transitional prompt representation, formulated as $M \in \mathbb{R}^{B\times C\times N_M\times H\times W}$, where C represents the RGB channels and $N_M = 1$ is adopted in our implementation.

Unlike static perception frames, the generated pseudo-temporal transitional prompt is driven by the feature differences of the current bi-temporal images. Therefore, it provides a more adaptive temporal representation for different input samples and enhances the subsequent video encoder's ability to model latent change processes.

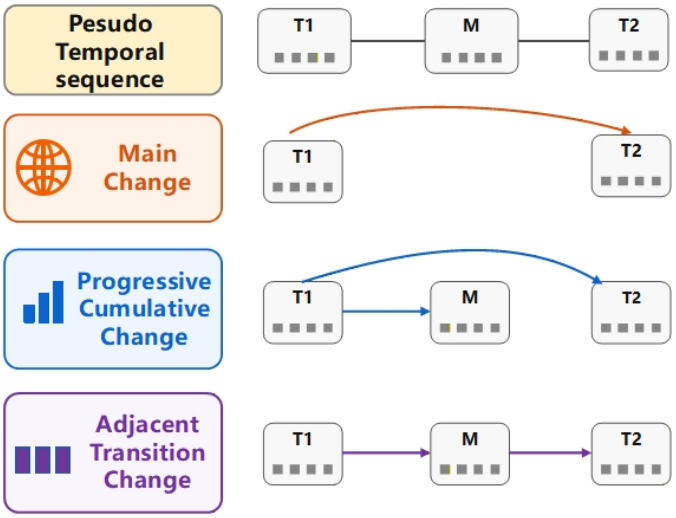

Fig. 2. Illustration of the multi-branch temporal difference decoding module.

C. Spatio-Temporal Feature Encoding

After obtaining the pseudo-temporal transitional prompt, this paper combines the pre-change image, the generated prompt representation, and the post-change image to form a short prompt-enhanced pseudo-temporal sequence:

$$X = [T_1, M, T_2]. \tag{6}$$

Here, M denotes the generated pseudo-temporal transitional prompt representation. Therefore, the length of the pseudo-temporal sequence is fixed as $T = 3$. Subsequently, the short sequence X is input into the video encoder to extract multi-scale spatio-temporal features:

$$\{F^\ell\}_{\ell=1}^{L} = \text{VideoEncoder}(X), \tag{7}$$

where L denotes the number of feature scales, and $F^\ell \in \mathbb{R}^{B\times C_\ell\times 3\times H_\ell\times W_\ell}$ represents the spatio-temporal feature at the ℓth scale. Here, B is the batch size, while C_ℓ, H_ℓ, and W_ℓ represent the channel number, height, and width of the feature at that scale, respectively.

Compared with encoding dual-temporal images separately, the video encoder can jointly model contextual relationships among the pre-change image, the generated transitional prompt

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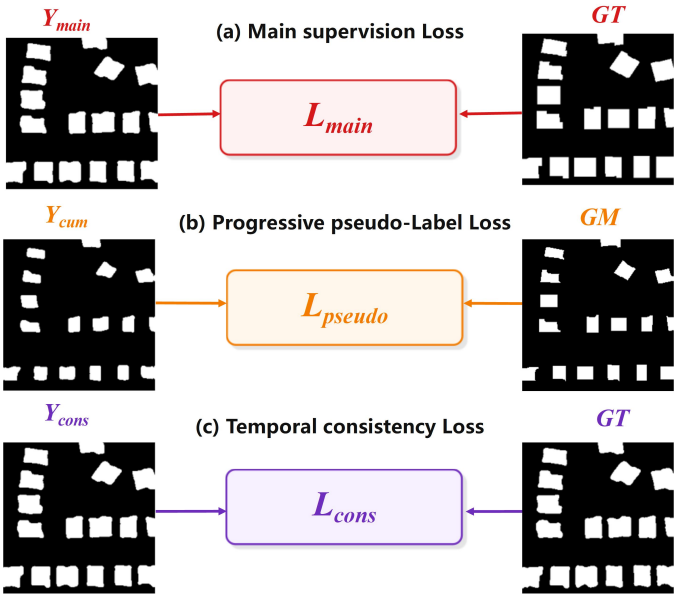

Fig. 3. Illustration of the proposed mixed pseudo-temporal transitional loss, which consists of three complementary supervision terms: the main supervision loss $\mathcal{L}_{main}$, the progressive pseudo-label loss $\mathcal{L}_{pseudo}$, and the temporal consistency loss $\mathcal{L}_{cons}$.

representation, and the post-change image in both temporal and spatial dimensions. In this way, the model obtains multi-scale features that contain richer information about the latent change process [40], [41]. In this paper, we adopt a Kinetics-400-pretrained X3D-L backbone as the video encoder to balance computational overhead and spatio-temporal modeling capability [40]. During training, the X3D-L encoder is fine-tuned rather than frozen to adapt the extracted spatio-temporal representations to the remote sensing change detection task.

D. Multi-Branch Temporal Decoding Module

A single direct difference between bi-temporal images is insufficient to fully characterize the latent change process, especially under few-shot supervision. To address this issue, we propose a Multi-Branch Temporal Decoding Module (MTDM) to decode complementary temporal differences from the constructed pseudo-temporal sequence. Specifically, MTDM consists of three branches, namely the main change branch, the progressive cumulative change branch, and the adjacent transition change branch. These branches share the same difference decoder, which enables temporal difference modeling without introducing a large number of additional parameters.

Given the multi-scale spatio-temporal features extracted by the video encoder, we denote the feature slice at the ℓth scale and time step t as F_t^ℓ. Since the pseudo-temporal sequence is composed of the pre-change image, the generated transitional prompt representation, and the post-change image, the temporal index t ranges from 0 to 2. Let $D(\cdot)$ denote the shared difference decoder for multi-scale feature decoding. The detailed structure of the shared temporal difference decoder is illustrated in Fig. 1. It adopts a multi-scale feature fusion strategy, where features from different scales are progressively upsampled and fused through skip connections, followed by a final upsampling layer and sigmoid activation to generate the change prediction map.

The main change branch aims to capture the overall change between the pre-change and post-change images. It decodes the endpoint temporal difference across all feature scales as:

$$Y_{\text{main}} = D\left(\{F_2^\ell - F_0^\ell\}_{\ell=1}^{L}\right), \tag{8}$$

where Y_{main} denotes the main change prediction map.

The progressive cumulative change branch is designed to describe the intermediate change state from the initial observation to the generated transitional prompt representation. Its prediction is obtained by decoding the cumulative temporal difference between the generated transitional prompt representation and the pre-change image:

$$Y_{\text{cum}} = D\left(\{F_1^\ell - F_0^\ell\}_{\ell=1}^{L}\right), \tag{9}$$

where Y_{cum} denotes the progressive cumulative change prediction.

The adjacent transition change branch focuses on the local temporal transition from the generated transitional prompt representation to the post-change image. Its prediction is obtained by decoding the adjacent temporal difference between the post-change image and the generated transitional prompt representation:

$$Y_{\text{trans}} = D\left(\{F_2^\ell - F_1^\ell\}_{\ell=1}^{L}\right). \tag{10}$$

To aggregate local transition responses, pixel-wise maximum fusion is applied to the two adjacent predictions:

$$Y_{\text{cons}}(u,v) = \max\left(Y_{\text{cum}}(u,v), Y_{\text{trans}}(u,v)\right), \tag{11}$$

where (u,v) denotes the spatial position. This operation preserves the stronger change response from the two pseudo-temporal transition stages at each pixel for subsequent temporal consistency supervision.

Through this design, the main change branch provides a stable endpoint change prediction, the progressive cumulative change branch introduces intermediate-stage temporal supervision, and the adjacent transition change branch enhances the sensitivity to local temporal variations. The three branches complement each other and improve the representation of latent change processes under few-shot conditions.

E. Mixed Pseudo-Temporal Transitional Loss

The proposed framework is optimized by a mixed pseudo-temporal transitional loss, which combines the main supervision loss, the progressive pseudo-label loss, and the temporal consistency loss, as shown in Fig. 3. Let GT denote the ground-truth change label. The main supervision loss constrains the prediction of the main change branch and is defined as:

$$\mathcal{L}_{\text{main}} = \mathcal{L}_{\text{BCE}}(Y_{\text{main}}, GT) + \mathcal{L}_{\text{Dice}}(Y_{\text{main}}, GT). \tag{12}$$

To supervise intermediate-stage predictions, this paper first derives a partial change label from the ground-truth change label GT by retaining 50% of the changed pixels. The partial

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change label is then expanded through morphological dilation to construct the pseudo-label GM, which provides progressive supervision for the cumulative change prediction. The progressive pseudo-label loss is applied to the output of the progressive cumulative change branch:

$$\mathcal{L}_{\text{pseudo}} = \mathcal{L}_{\text{BCE}}(Y_{\text{cum}}, GM). \tag{13}$$

In addition, a temporal consistency loss is introduced to directly supervise the aggregated adjacent transition prediction using the ground-truth change label, encouraging the pseudo-temporal transition response to be consistent with the actual changed regions:

$$\mathcal{L}_{\text{cons}} = \mathcal{L}_{\text{BCE}}(Y_{\text{cons}}, GT) + \mathcal{L}_{\text{Dice}}(Y_{\text{cons}}, GT). \tag{14}$$

The overall training objective is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \alpha_1 \mathcal{L}_{\text{pseudo}} + \alpha_2 \mathcal{L}_{\text{cons}}, \tag{15}$$

where α_1 and α_2 are weighting coefficients for the progressive pseudo-label loss and temporal consistency loss, respectively.

Overall, the proposed framework first employs the Dynamic Pseudo-Temporal Prompt Generation Module (DPGM) to generate a sample-specific pseudo-temporal transitional prompt M according to the feature differences between bi-temporal images, which captures the latent transition information between the pre-change and post-change observations. Subsequently, the pseudo-temporal sequence containing the generated prompt is fed into the pre-trained video encoder to extract spatio-temporal feature representations with latent temporal transition cues. Based on these features, the Multi-Branch Temporal Decoding Module (MTDM) decodes complementary temporal change information from three perspectives, including main change, progressive cumulative change, and adjacent transition change. Finally, the mixed pseudo-temporal transitional loss jointly supervises and optimizes the predictions from different branches, enabling the model to learn more reliable temporal change representations under few-shot supervision.

IV. Experimental Results and Analysis

A. Datasets

We conduct experiments on three public datasets, including LEVIR-CD, WHU-CD, and CLCD, to validate the effectiveness of the proposed method. Specifically, LEVIR-CD and WHU-CD mainly focus on building change detection, whereas CLCD is designed for cropland change detection. The main characteristics of these datasets are summarized in Table I.

1) LEVIR-CD [10]. The LEVIR-CD dataset was collected from Google Earth images covering multiple urban areas in Texas, USA, and is widely used for high-resolution building change detection. It provides annotations for building changes, including newly constructed and demolished buildings. The dataset contains 637 pairs of bi-temporal images with a size of 1024×1024 and a spatial resolution of 0.5 m, and is divided into a training set (445 pairs), a validation set (64 pairs), and a testing set (128 pairs).

2) WHU-CD [42]. The WHU-CD dataset was constructed from high-resolution aerial images of Christchurch, New Zealand, acquired in 2012 and 2016. It mainly focuses on building changes caused by post-earthquake reconstruction. Following the commonly used 256×256 cropping protocol, the dataset is divided into a training set (6096 pairs), a validation set (762 pairs), and a testing set (762 pairs).

TABLE I
Summary of Dataset Characteristics

Dataset	Image Size	Resolution	Pairs (Train/Val/Test)
LEVIR-CD	1024×1024	0.5 m	445/64/128
WHU-CD	256×256	0.2 m	6096/762/762
CLCD	512×512	0.5–2 m	360/120/120

3) CLCD [43]. The CLCD dataset was built from Gaofen-2 images acquired in Guangdong Province, China, in 2017 and 2019, with the goal of cropland change detection. It consists of 600 pairs of bi-temporal images with a size of 512×512 and a spatial resolution ranging from 0.5 m to 2 m, and is divided into a training set (360 pairs), a validation set (120 pairs), and a testing set (120 pairs).

B. Experimental Settings

To ensure a consistent and reproducible experimental protocol, we follow the official or commonly used training/validation/test splits for each dataset. All input images and corresponding labels are cropped into 256×256 patches for model training and evaluation. Under the limited-data setting, we construct different training subsets by randomly sampling labeled samples from the training set. Specifically, the 10-image and 50-image settings are generated by randomly selecting 10 and 50 labeled image pairs, respectively. Moreover, to further investigate the model performance under different annotation ratios, percentage-based subsets containing 5% and 15% of the training samples are also constructed. For each limited-data setting, the random sampling process is independently repeated five times, and the average performance is reported to reduce the impact of sampling randomness. Since the supervision information is extremely limited in the 10-image setting, all selected samples are required to contain changed regions to provide effective supervision for change detection. During training, data augmentation strategies, including random cropping, scaling, random horizontal and vertical flipping, and random temporal swapping, are adopted. For fair comparisons, all competing methods are trained using the same limited-data subsets and evaluated on the same validation and test sets.

All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 4090 GPU. The model is trained using the Adam optimizer with an initial learning rate of 1×10^{-3}, a weight decay of 1×10^{-4}, $\beta_1 = 0.9$, and $\beta_2 = 0.99$. The number of training epochs is set to 200, the batch size is set to 8, and a polynomial learning rate decay strategy is adopted. Based on the ablation study, both hyperparameters α_1 and α_2 in the mixed pseudo-temporal transitional loss are set to 1. For each limited-data setting, five different training subsets are independently sampled from the original training set. The sampling process is performed independently, while

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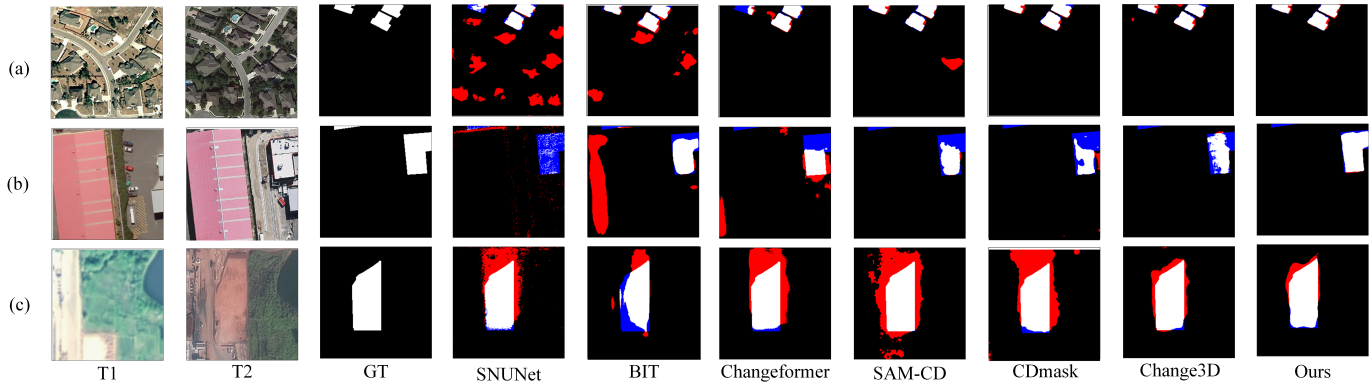

Fig. 4. Visualization results of different methods on (a) LEVIR-CD, (b) WHU-CD, and (c) CLCD. Red indicates false positives, and blue indicates false negatives.

the training random seed is fixed to 34 for all experiments to ensure fair and reproducible model optimization.

C. Evaluation Metrics

We adopt IoU_1, F1, P_1, and R_1 to evaluate change detection performance [44]. The metrics are defined as:

$$P_1 = \frac{TP}{TP+FP}, \qquad R_1 = \frac{TP}{TP+FN}, \\ \mathrm{F1} = \frac{2P_1R_1}{P_1+R_1}, \qquad \mathrm{IoU}_1 = \frac{TP}{TP+FP+FN}. \tag{16}$$

Here, TP, FP, and FN denote the numbers of true positive, false positive, and false negative pixels of the changed class, respectively. P_1 and R_1 reflect false detections and missed detections, respectively, while F1 and IoU_1 measure the overall prediction quality of changed regions. Higher values indicate better performance [11].

D. Comparative Experiments

To validate the effectiveness of the proposed method, we compare it with six representative change detection methods, including SNUNet [5], BIT [15], ChangeFormer [12], SAM-CD [17], CDmask [45], and Change3D [31]. Specifically, SNUNet is a CNN-based method that enhances multi-scale feature fusion through dense connections; BIT and ChangeFormer are Transformer-based methods for modeling global contextual relationships and multi-scale change representations; SAM-CD introduces the visual foundation model SAM into change detection; CDmask models change regions from a mask-based perspective; and Change3D reformulates bi-temporal change detection from a video modeling perspective. For fair comparison, all methods are trained using the same limited training sample splits and evaluated under the same validation and test settings.

Table II reports the comparison results under the extremely few-shot settings with 10 and 50 training images. As shown in the table, our method achieves the best performance on most metrics across the three datasets, particularly in terms of the primary IoU and F1 metrics. In particular, it maintains stable change detection performance with only 10 labeled samples, indicating that the proposed Dynamic Pseudo-Temporal Prompt Generation Module and Multi-Branch Temporal Decoding Module can effectively exploit latent change information within bi-temporal images under extremely limited supervision. In comparison, SNUNet performs relatively weakly, suggesting that local convolutional features and multi-scale fusion are insufficient for complex change modeling in extreme few-shot scenarios. BIT and ChangeFormer benefit from more training samples but degrade clearly in the 10-sample setting, while SAM-CD shows unstable performance across different datasets, indicating that general segmentation priors may not be consistently transferable to few-shot change detection. CDmask achieves competitive results in some settings, but its IoU and F1 scores remain lower than those of our method across the three datasets, indicating that its mask-based modeling strategy is less effective under extremely limited supervision. Change3D obtains strong results, demonstrating

TABLE II
COMPARISON RESULTS UNDER 10-IMAGE AND 50-IMAGE SETTINGS

Dataset	Model	10 images				50 images			
		IoU_1	F1	P_1	R_1	IoU_1	F1	P_1	R_1
LEVIR-CD	BIT	21.64	33.51	48.02	27.59	51.99	68.41	77.07	61.71
	ChangeFormer	18.56	30.98	44.36	24.17	42.46	59.58	66.72	53.96
	SNUNet	3.40	5.37	5.25	8.78	11.82	18.22	17.36	26.20
	SAM-CD	19.41	30.79	54.51	21.70	50.15	66.42	68.75	66.10
	CDmask	25.67	40.63	52.08	34.37	54.81	70.79	78.76	64.66
	Change3D	49.04	65.73	72.49	60.64	69.90	82.28	79.73	**85.10**
	Ours	**60.52**	**75.36**	**77.74**	**73.30**	**74.07**	**85.10**	**88.56**	81.97
WHU-CD	BIT	20.16	33.52	24.59	58.71	26.77	42.03	36.12	55.28
	ChangeFormer	25.69	40.76	39.01	45.17	45.09	62.15	65.60	59.15
	SNUNet	8.07	13.36	10.23	28.35	22.94	34.56	27.77	57.40
	SAM-CD	6.42	12.05	7.45	31.91	15.99	27.14	20.45	45.72
	CDmask	36.99	54.00	43.94	70.03	42.34	59.50	45.75	**85.06**
	Change3D	53.88	69.93	70.17	70.38	69.40	81.92	84.35	79.66
	Ours	**59.93**	**74.81**	**75.31**	**74.52**	**72.50**	**84.05**	**85.98**	82.30
CLCD	BIT	18.73	29.76	46.97	25.88	31.89	48.00	51.08	45.65
	ChangeFormer	11.99	20.89	45.57	15.07	20.75	34.19	48.62	26.69
	SNUNet	11.56	18.54	20.23	23.79	18.63	28.86	34.85	28.81
	SAM-CD	5.42	8.54	22.13	11.56	29.50	45.31	**65.14**	35.38
	CDmask	19.04	31.98	47.73	22.61	26.83	42.31	63.83	31.36
	Change3D	19.33	32.40	**48.42**	24.35	39.86	56.99	57.06	56.96
	Ours	**24.02**	**38.26**	41.91	**37.51**	**41.46**	**58.58**	58.09	**60.01**

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TABLE III
COMPARISON RESULTS UNDER 5% AND 15% TRAINING SETTINGS

Dataset	Model	5%				15%			
		IoU_1	F1	P_1	R_1	IoU_1	F1	P_1	R_1
LEVIR-CD	BIT	65.25	78.97	83.01	75.38	73.91	84.99	87.53	82.62
	ChangeFormer	62.38	76.83	81.71	72.61	73.94	85.02	88.22	82.04
	SNUNet	33.39	48.56	50.08	49.99	36.37	52.08	54.44	52.30
	SAM-CD	70.39	82.62	86.68	78.95	76.32	86.57	88.02	85.16
	CDmask	76.04	86.39	85.08	87.14	79.01	88.30	**93.84**	83.38
	Change3D	77.87	87.55	89.51	85.70	81.58	89.86	91.12	88.63
	Ours	**79.38**	**88.50**	**89.85**	**87.20**	**81.88**	**90.04**	91.06	**89.04**
WHU-CD	BIT	44.05	60.75	53.31	74.92	55.77	71.40	68.82	75.02
	ChangeFormer	56.64	72.30	80.33	65.84	63.67	77.76	86.71	70.53
	SNUNet	40.36	55.12	48.12	71.55	50.38	64.94	61.50	73.78
	SAM-CD	58.90	73.84	89.48	63.70	74.54	85.41	87.93	83.08
	CDmask	57.15	72.73	63.77	84.62	65.42	79.10	71.24	**88.90**
	Change3D	78.40	87.98	90.11	85.78	81.03	89.52	93.32	86.02
	Ours	**79.83**	**88.78**	**90.24**	**87.41**	**82.24**	**90.25**	**93.75**	87.05
CLCD	BIT	23.78	38.18	50.11	33.61	29.48	44.49	58.64	38.13
	ChangeFormer	29.51	45.51	56.34	38.81	35.67	52.56	53.61	51.78
	SNUNet	22.02	33.68	34.83	39.60	28.76	42.36	43.05	47.29
	SAM-CD	33.66	50.17	65.16	42.44	42.98	60.04	68.54	53.97
	CDmask	31.01	47.34	**68.26**	36.24	44.90	61.98	67.42	57.35
	Change3D	42.29	59.39	60.54	**58.55**	49.51	66.21	71.58	61.74
	Ours	**45.27**	**62.23**	67.21	58.02	**51.48**	**67.94**	**73.98**	**62.81**

the usefulness of video-inspired temporal modeling. However, its static perception frames are less adaptive to sample-specific change content. By contrast, our method explores the latent change process through dynamic pseudo-temporal transitional prompts, leading to better training stability and change recognition.

Table III further presents the comparison results using 5% and 15% of the training samples. With more training samples, all methods generally achieve improved performance. Nevertheless, our method still obtains the best or highly competitive results on most metrics across the three datasets, especially on IoU and F1, showing its effectiveness under both extremely few-shot and general limited-annotation settings. SNUNet remains relatively weak, while BIT and ChangeFormer improve with increased annotations but still lag behind our method in complex scenes. SAM-CD performs well in some cases but fluctuates across datasets and settings, suggesting that foundation-model priors do not always provide stable advantages under limited annotations. CDmask also achieves competitive performance under these settings, whereas our method maintains consistently higher IoU and F1 scores across all three datasets, demonstrating more reliable overall performance under limited supervision. Change3D achieves strong performance, further confirming the benefit of temporal modeling. In contrast, our method further mines the latent change process within each bi-temporal sample through dynamic pseudo-temporal transitional prompts and multi-branch temporal decoding, resulting in more stable change detection performance under low annotation ratios.

Fig. 4 presents qualitative results under the 5% training sample setting. SNUNet tends to miss many changed regions. BIT and ChangeFormer can detect part of the changed areas, but still suffer from boundary errors and local missed detections. SAM-CD can locate major changed objects in some cases, but often produces large false positive regions. CDmask can capture the main changed regions relatively well, but still exhibits some false positives and missed detections around object boundaries and irregular regions. Change3D produces results closer to the ground truth overall, but still has false positives and false negatives around small boundaries and irregular regions. In contrast, the proposed method generates more complete and cleaner change maps with noticeably fewer false positives and false negatives, further validating its ability to identify changed regions under limited data.

E. Ablation Study

To evaluate the contribution of each proposed component, we conduct ablation experiments on the LEVIR-CD dataset under the 10-image training setting, and the results are reported in Table IV. DPGM denotes the Dynamic Pseudo-Temporal Prompt Generation Module, PCCB denotes the progressive cumulative change branch, and ATCB denotes the adjacent transition change branch. In the ablation experiments, both the network architecture and the corresponding optimization objective are gradually changed with the introduction of different branches. Specifically, the baseline and the DPGM-only variant are optimized using the main supervision loss L_{main}. After introducing PCCB, the progressive cumulative change prediction branch is additionally optimized with the pseudo-temporal supervision loss L_{pseudo}. With the further introduction of ATCB, the adjacent transition prediction branch is constrained by the temporal consistency loss L_{cons}. Therefore, Table IV reflects the effects of different temporal modeling branches and their corresponding loss components on the model performance.

1) Effect of the Dynamic Pseudo-Temporal Prompt Generation Module (DPGM): Compared with the baseline, adding DPGM improves IoU_1 and F1 by 8.48 and 7.29 percentage points, respectively. Meanwhile, R_1 increases by 11.46 percentage points. This indicates that the generated pseudo-temporal transitional prompts provide effective transition cues between the pre-change and post-change images, enabling the model to detect more changed regions under extremely limited supervision.

2) Effect of the Progressive Cumulative Change Branch (PCCB): Based on DPGM, introducing PCCB further improves IoU_1 and F1 by 2.45 and 1.92 percentage points, respectively. In this setting, the progressive cumulative change prediction branch is optimized with the pseudo-temporal supervision loss L_{pseudo}, which provides additional supervision from the generated pseudo-temporal transition process. More importantly, P_1 increases from 74.10% to 80.29%, achieving the highest value among all variants. This indicates that the progressive cumulative change branch combined with pseudo-temporal supervision helps the model focus on more reliable changed regions and reduce false positives, although R_1 slightly decreases due to its more conservative prediction tendency.

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TABLE IV
ABLATION STUDY OF DIFFERENT MODULES AND OPTIMIZATION OBJECTIVES ON THE LEVIR-CD DATASET

Setting	DPGM	PCCB	ATCB	Loss Function	IoU_1	F1	$\mathrm{Precision}_1$	Recall_1
Baseline	×	×	×	L_{main}	49.04	65.73	72.49	60.64
Baseline + DPGM	✓	×	×	L_{main}	57.52	73.02	74.10	72.10
Baseline + DPGM + PCCB	✓	✓	×	$L_{main} + L_{pseudo}$	59.97	74.94	**80.29**	70.44
Baseline + DPGM + ATCB	✓	×	✓	$L_{main} + L_{cons}$	59.72	74.69	76.38	73.23
Baseline + DPGM + PCCB + ATCB	✓	✓	✓	$L_{main} + L_{pseudo} + L_{cons}$	**60.52**	**75.36**	77.74	**73.30**

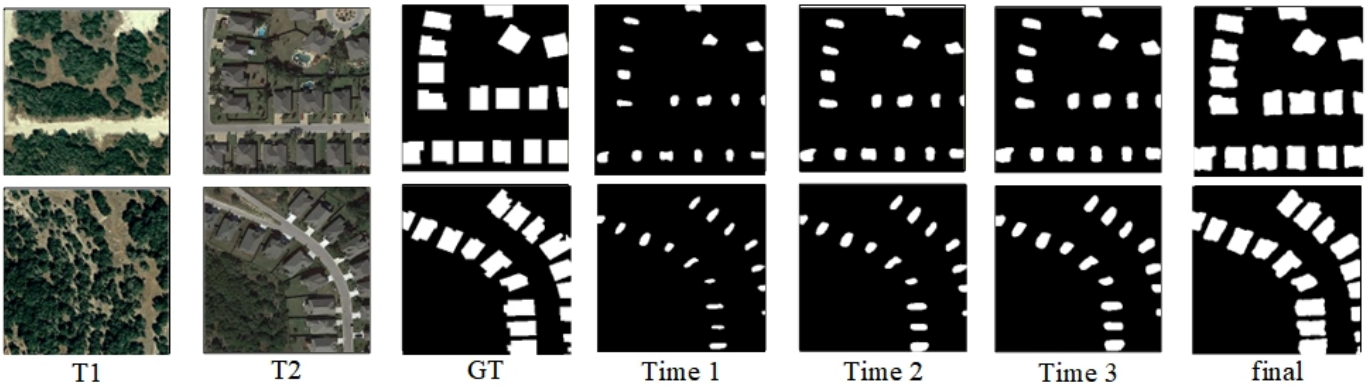

Fig. 5. Visualization of stage-wise predictions with three generated pseudo-temporal transitional prompts under the full LEVIR-CD training set.

3) Effect of the Adjacent Transition Change Branch (ATCB): Adding ATCB on top of DPGM also brings consistent improvements, increasing IoU_1 and F1 by 2.20 and 1.67 percentage points, respectively. In this setting, the aggregated adjacent transition prediction is directly supervised by the ground-truth change label through the temporal consistency loss $\mathcal{L}_{\mathrm{cons}}$, which encourages the pseudo-temporal transition response to be consistent with the actual changed regions. Compared with PCCB, ATCB achieves a higher R_1, indicating that adjacent transition modeling with temporal consistency constraint is more effective in capturing local temporal changes and recovering changed regions that may be missed by cumulative change modeling alone.

When DPGM, PCCB, and ATCB are jointly used, the full model achieves the best overall performance. Compared with the baseline, IoU_1 and F1 are improved by 11.48 and 9.63 percentage points, respectively. These results demonstrate that PCCB and ATCB provide complementary temporal modeling cues. PCCB improves the reliability of change prediction through pseudo-temporal supervision, while ATCB enhances the ability to recover changed regions through temporal consistency constraint. Their combination enables the model to better exploit the latent pseudo-temporal change process and achieve more stable performance under the extremely few-shot setting.

F. Discussion

1) Effect of the Number of Pseudo-Temporal Transitional Prompts: To analyze the influence of the number of pseudo-temporal transitional prompts, we conduct experiments on the LEVIR-CD dataset under limited-data settings by varying the number of generated pseudo-temporal transitional prompts from 1 to 7. As shown in Table V, using only one pseudo-temporal transitional prompt achieves the best overall performance in both the 10-image and 50-image settings. As the number of pseudo-temporal transitional prompts increases, the model performance does not improve consistently, but instead shows an overall downward trend. This indicates that, when supervised samples are limited, an overly long pseudo-temporal sequence may increase the optimization difficulty and make the model more likely to fit unreliable transitional representations.

TABLE V
INFLUENCE OF DIFFERENT NUMBERS OF PSEUDO-TEMPORAL TRANSITIONAL PROMPTS ON THE LEVIR-CD DATASET

Prompts	10 images				50 images			
	IoU_1	F1	P_1	R_1	IoU_1	F1	P_1	R_1
1	**60.52**	**75.36**	**77.74**	**73.30**	**74.07**	**85.10**	**88.55**	81.97
2	54.39	70.36	70.12	71.34	72.35	83.95	86.29	81.79
3	53.85	69.93	69.75	70.43	71.04	83.05	85.60	80.68
4	53.67	69.73	69.37	70.36	70.29	82.54	83.43	81.75
5	55.79	71.54	72.50	70.94	71.29	83.23	84.53	**81.98**
6	52.54	68.82	71.43	66.75	70.84	82.91	85.47	80.59
7	50.89	67.38	66.07	69.01	70.16	82.44	84.52	80.50

Although multiple pseudo-temporal transitional prompts are not the optimal choice under limited-data settings, they can still provide meaningful stage-wise change information when sufficient supervision is available. To verify this, we additionally train a model variant with three generated pseudo-temporal transitional prompts on the full LEVIR-CD training set and visualize its stage-wise predictions in Fig. 5. As the pseudo-temporal sequence progresses from the initial time to

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TABLE VI
COMPARISON OF DIFFERENT PSEUDO-TEMPORAL PROMPT GENERATION STRATEGIES ON THE LEVIR-CD DATASET

Strategy	IoU_1	F1	P_1	R_1
Random	56.68	72.31	72.03	72.94
Gaussian	57.16	72.71	73.02	72.50
Linear Interpolation	58.52	73.77	75.26	72.99
Feature Interpolation	59.97	74.90	77.34	72.74
Learnable Frame	59.77	74.78	77.68	72.08
Ours	**60.52**	**75.36**	**77.74**	**73.30**

TABLE VII
COMPARISON OF DIFFERENT PSEUDO-LABEL GENERATION STRATEGIES ON THE LEVIR-CD DATASET

Strategy	IoU_1	F1	P_1	R_1
Random generation	59.75	74.78	75.90	**74.06**
Morphological dilation	**60.52**	**75.36**	**77.74**	73.30

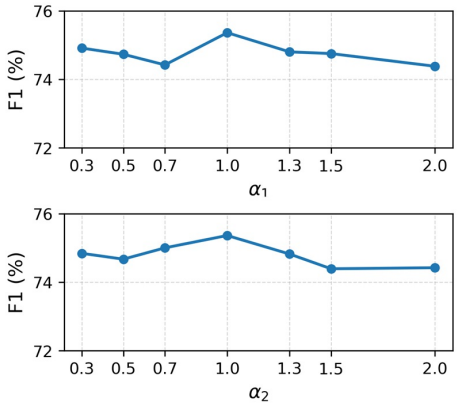

Fig. 6. Sensitivity analysis of F1 score with respect to different hyperparameters (α_1 and α_2).

TABLE VIII
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS

Method	Params (M)	FLOPs (G)	Time (ms)	FPS
SNUNet	12.0350	46.6968	8.0017	124.97
BIT	12.4025	10.8718	7.6205	131.23
ChangeFormer	29.7480	17.5714	12.7710	78.30
SAM-CD	2.5914	43.2200	25.6255	39.02
Change3D	6.4246	7.3624	10.4139	96.03
Ours	6.4640	23.7411	14.6719	68.16

the final time, the predicted changed regions gradually evolve from weak and local responses to more complete structures. This demonstrates that, with sufficient training data, multi-prompt pseudo-temporal modeling can help the model learn progressive change processes. This analysis is used to illustrate the potential of progressive pseudo-temporal modeling, while the one-prompt setting is adopted as the default configuration in all limited-data experiments.

2) Comparison of Transitional Prompt Generation and Pseudo-Label Generation Strategies: To investigate the effectiveness of different pseudo-temporal prompt generation strategies, we compare the proposed dynamic prompt generation strategy with several alternative construction methods, including random generation, Gaussian generation, linear interpolation, feature interpolation, and static learnable frame. As shown in Table VI, the proposed dynamic pseudo-temporal prompt generation strategy achieves the best overall performance, obtaining the highest IoU_1 and $\mathrm{F1}$ scores among all compared methods. Compared with random and Gaussian generation, the proposed strategy generates transition prompts based on the corresponding bi-temporal image content rather than unconstrained perturbations. Moreover, compared with interpolation-based and static learnable alternatives, the dynamic strategy provides sample-adaptive temporal transition information, demonstrating its effectiveness in modeling the latent transition process between bi-temporal observations.

We further compare the influence of different pseudo-label generation strategies, including random generation and morphological dilation. As shown in Table VII, the morphological dilation strategy achieves better overall performance in terms of IoU_1 and $\mathrm{F1}$. This indicates that pseudo-labels generated by morphological dilation can better reflect the progressive temporal change process and provide more effective supervision for progressive-stage predictions.

3) Hyper-parameter Analysis: We further analyze the influence of the loss hyper-parameters α_1 and α_2. Specifically, we vary each hyper-parameter from 0.3 to 2.0 and report the corresponding F1 scores. As shown in Fig. 6, the performance remains stable under different settings of both α_1 and α_2, with only slight fluctuations in F1. This indicates that the proposed method is not sensitive to these two loss hyper-parameters and can maintain stable performance within a reasonable range. Since the best performance is achieved when $\alpha_1 = 1.0$ and $\alpha_2 = 1.0$, we set both hyper-parameters to 1.0 in our experiments.

4) Computational Efficiency: Table VIII compares the computational efficiency of different methods. The proposed method has 6.464M parameters, which is comparable to Change3D and significantly lower than SNUNet, BIT, and ChangeFormer. This indicates that the proposed dynamic pseudo-temporal prompt generation and multi-branch temporal decoding do not substantially increase model size. Because pseudo-temporal transitional prompt modeling and multi-branch difference decoding introduce additional computation, the FLOPs of the proposed method are higher than those of Change3D, but still lower than those of SNUNet and SAM-CD.

In terms of practical inference speed, the proposed method reaches 68.16 FPS. Although it is slower than the lightweight Change3D, it is clearly faster than SAM-CD and maintains acceptable inference efficiency. Combined with the detection results in Table II, the proposed method achieves more stable limited-data change detection performance with moderate computational cost, balancing model complexity, inference efficiency, and detection accuracy.

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V. Conclusion

This paper proposed a dynamic pseudo-temporal prompt generation-enhanced framework for few-shot remote sensing change detection. By generating sample-specific pseudo-temporal transitional prompts, the proposed method mines latent transition cues from bi-temporal samples without relying on explicit pixel-level intermediate image generation. Furthermore, the framework integrates pretrained video-based spatio-temporal encoding, multi-branch temporal decoding, progressive pseudo-label supervision, and temporal consistency learning to enhance change representation learning under limited annotations. Experiments on LEVIR-CD, WHU-CD, and CLCD demonstrate that the proposed method consistently outperforms representative change detection methods across multiple few-shot settings. Ablation studies further verify the effectiveness of the proposed prompt generation module and the multi-branch temporal decoding design. In future work, we will explore more robust pseudo-temporal transition modeling strategies for complex few-shot remote sensing change detection scenarios.

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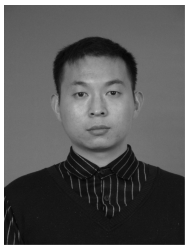

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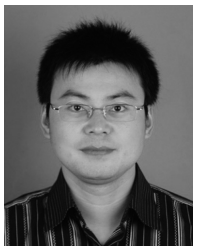

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